

Stingray™ PS225 SmartNIC Adapters

Dual-Port 25 Gb/s PCI Express Ethernet SmartNIC Adapters

General Description

The Broadcom® Stingray™ PS225 family of SmartNIC adapters, based on Broadcom's latest data center SoC technology, offer groundbreaking network and compute power in a compact package. The PS225 adapters are PCI Express Low-Profile, half-length adapters, and provide two 25-Gigabit Ethernet ports along with eight 64-bit ARMv8 processors running at an unmatched 3.0 GHz clock rate. The adapters are available in a range of onboard DDR4 memory sizes, including 4 GB, 8 GB, and 16 GB.

Features

- **PCIe Interface:**
 - x8 PCI Express v3.0 compliant
 - Single-Root I/O Virtualization (SR-IOV) with up to 128 VFs
 - Function Level Reset (FLR) support
- **Processing Subsystem:**
 - 8-core ARMv8 A72 64-bit processors at 3.0 GHz
 - 16 MB total cache (8 MB L2 + 8 MB L3)
 - Two 64-bit channels of DDR4 memory at 2,400 MT/s
- **Hardware Accelerators:**
 - TruFlow™ configurable flow accelerator engine
 - Line-rate Crypto engine with single-pass hashing and encryption
 - RAID 5/6 engine
- **Network Interface**
 - Dual-port SFP28 pluggable media interface, supporting 25GbE or 10GbE optical transceiver or direct-attach copper (DAC) cable
 - RDMA over Converged Ethernet (RoCE) v1 and v2
 - Advanced Congestion Avoidance
 - Virtual Network Termination – VXLAN, NVGRE, Geneve, GRE
 - Multiqueue, NetQueue, and VMQ
 - Tunnel-aware stateless offloads
 - IPv4 and IPv6
 - TCP, UDP, and IP checksum

- Large Send Offload (LSO)
- Large Receive Offload (LRO)
- TCP Segmentation Offload (TSO)
- Receive-side Scaling (RSS)
- Transmit-side Scaling (TSS)
- VLAN insertion/removal
- DCB support: PFC, ETS, QCN, DCBx
- Jumbo frames (up to 9 KB)
- Network boot (PXE, UEFI, iSCSI)
- **Security and Manageability**
 - Secure Boot
 - Secure Key Storage
 - ARM TrustZone PKA Engine
 - NC-SI over MCTP (SMBus, PCIe VDM)
 - NC-SI over RMB
- **Form Factor**
 - PCI Express low-profile, half-length
 - I/O brackets included for both low-profile and full-height slots

Applications

- High-performance data plane processing
- Vswitch-on-SmartNIC
- Network Function Virtualization (NFV)

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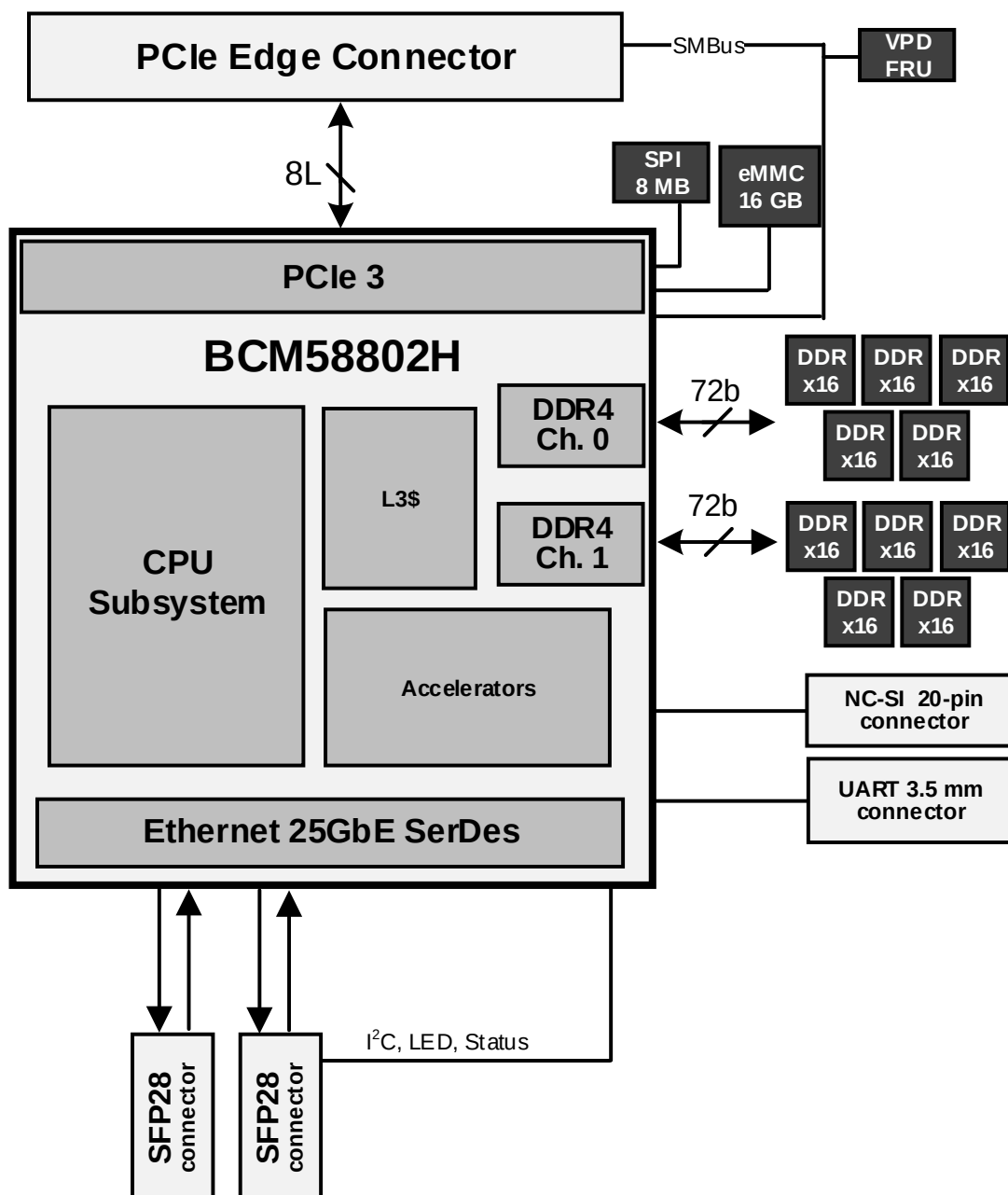
1 Functional Description

This section provides the functional description of the PS225 SmartNIC Adapter Cards.

1.1 Block diagram

Figure 1 shows the main functional blocks of the PS225 SmartNIC Adapter Cards.

Figure 1: Block Diagram



2 Connectors

2.1 Host Interface Connector

The PS225 Stingray SmartNIC adapters are PCI Express 3 x8 add-in cards. The card edge connector pinout complies with the PCI Express Card Electromechanical Specification, and the interface is Gen3 compliant.

NOTE: Please refer to the PCI Express Card Electromechanical Specification v3.0 for additional details on the card edge connector pinout.

2.1.1 SMBus Interface

The PS225 adapters connect the SMBus pins from the PCI Express edge connector to a 128 KB serial EEPROM and to the BCM58802H ASIC, to enable access by the system baseboard management controller (BMC). The BCM58802H device acts as SMBus slave.

2.2 Network Connectors

The PS225 adapters have two SFP28 connectors that are compliant to the SFF-8402 specification. The BCM58802H device has integrated PHYs that are tied directly to the SFP28 connectors. The SFP28 connectors can support passive or active direct-attach copper (DAC) cables as well as active optical modules that meet industry specifications for the interface. The adapter is capable of linking at either 25 Gb/s or 10 Gb/s for both connectors. Concurrent mixed connection speeds (25 Gb/s and 10 Gb/s) are not supported.

2.3 UART Connector

The PS225 adapters provide console access using a 3.5 mm audio jack connector on the I/O bracket, which is accessible external to the system. The connector supports a simple UART interface (TX, RX) at TTL signaling levels (+3.3V). The port does NOT support RS-232 or RS-485 signaling levels. The port must be accessed using a cable that supports TTL signaling level. The following cable assembly is commercially available and has been tested with the PS225 adapters:

- Manufacturer – FDTI
- Manufacturer Part Number – TTL-232R-3V3-AJ

2.4 NC-SI Connector

An NC-SI connector is provided for optional cable connection to the host system BMC, for sideband port access. The connector is a 20-pin header with 2 mm pitch, compatible with Tyco PN 3-1470209-3 with one pin removed as a keyway to enable proper connector orientation when plugging. The connector pinout is shown in [Table 1](#).

Table 1: NC-SI Connector Pinout

Pin	Signal	Signal	Pin
1	GND	NCSI_TXD1	2
3	GND	NCSI_TXENA	4
5	NCSI_RXD1	GND	6
7	REFCLK	GND	8
9	NCSI_TXD0	NCSI_CRSDV	10
11	GND	NCSI_RXD0	12

Table 1: NC-SI Connector Pinout (Continued)

Pin	Signal	Signal	Pin
13	RX_ER (optional)	Pin removed for keying	14
15	VAUX (supply)	VAUX (supply)	16
17	GND	+3.3V	18
19	VAUX (supply)	VAUX (supply)	20

3 BCM58802H ASIC

The PS225 adapters use the Broadcom BCM58802H ASIC. The BCM58802H ASIC integrates the latest Broadcom high-performance Network Interface Controller (NIC) in a system-on-chip (SoC) featuring an 8-core ARM Cortex-A72 processor subsystem.

Refer to BCM5880X ASIC documentation for further detail on the BCM58802H device.

4 Onboard Memories

4.1 Nitro SPI Flash (Non-Volatile)

The PS225 adapters have a 64-Mbit (8 MB) SPI-based flash memory that is used to store firmware for the embedded Nitro Ethernet NIC.

4.2 eMMC (Non-Volatile)

The PS225 adapters have a 128-Gbit (16 GB) eMMC Flash memory that is used to store the AP (ARM) bootloaders and the OS filesystem.

4.3 VPD EEPROM (Non-Volatile)

The PS225 adapters have a 128-kbit serial EEPROM attached to the SMBus interface on the PCIe connector that can be used to store vital product data (VPD).

4.4 DDR4 SDRAM (Volatile)

The PS225 adapters support DDR4 SDRAM using 10 discrete DDR4 components across two memory channels. The overall size of the memory array is different for each variant of the adapter, as shown in [Table 2](#).

Table 2: Adapter Variants

Adapter Variant	DRAM Devices	Total memory array size
PS225-H04	4 Gb (256 Mb × 16)	4 GB
PS225-H08	8 Gb (512 Mb × 16)	8 GB
PS225-H16	16 Gb (1 Gb × 16)	16 GB

5 LEDs and Reset Button

5.1 LEDs

The PS225 adapters support link speed and activity indicators for each port on the I/O bracket (see [Table 3](#)). See [I/O Bracket Diagram](#) for details on LED location.

Table 3: LED States and Descriptions

LED	State	Description
Activity	Off	No activity
	Green blinking	Port activity/traffic
Link	Off	No link
	Green	Linked at 25 Gb/s
	Yellow	Linked at 10 Gb/s or 1 Gb/s

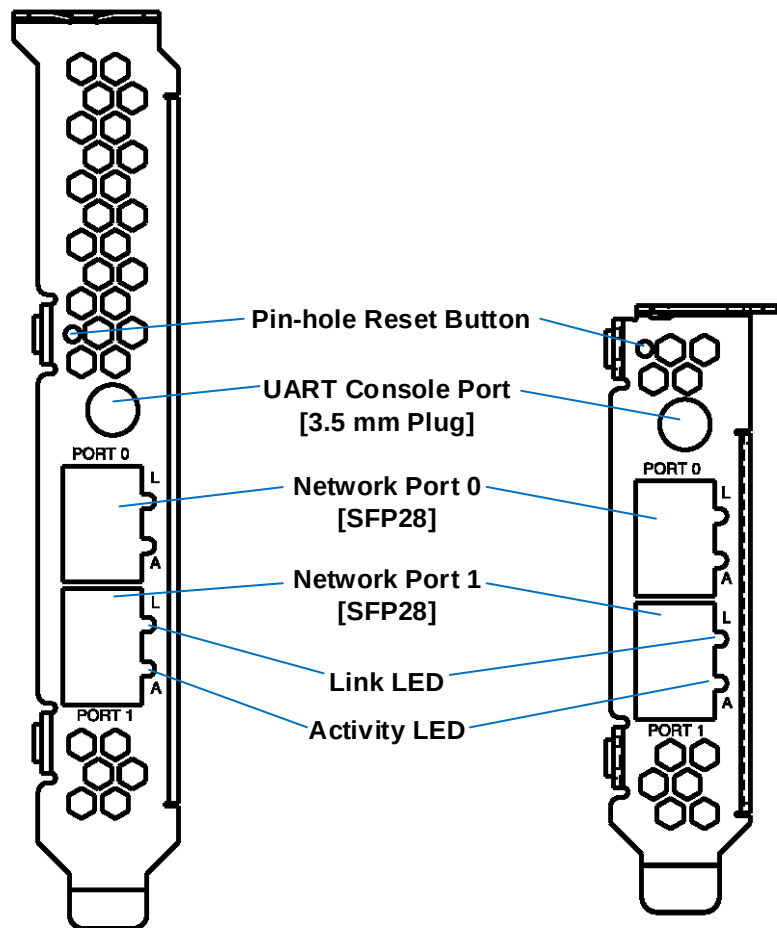
5.2 Reset Button

The PS225 adapters support a pin hole reset button, located at the I/O bracket. This reset button is debounced on the PS225 adapter and results in a global reset of the BCM58802H device.

6 I/O Bracket Diagram

Figure 2 describes the functions of the I/O bracket as viewed from outside the system.

Figure 2: PS225 Brackets

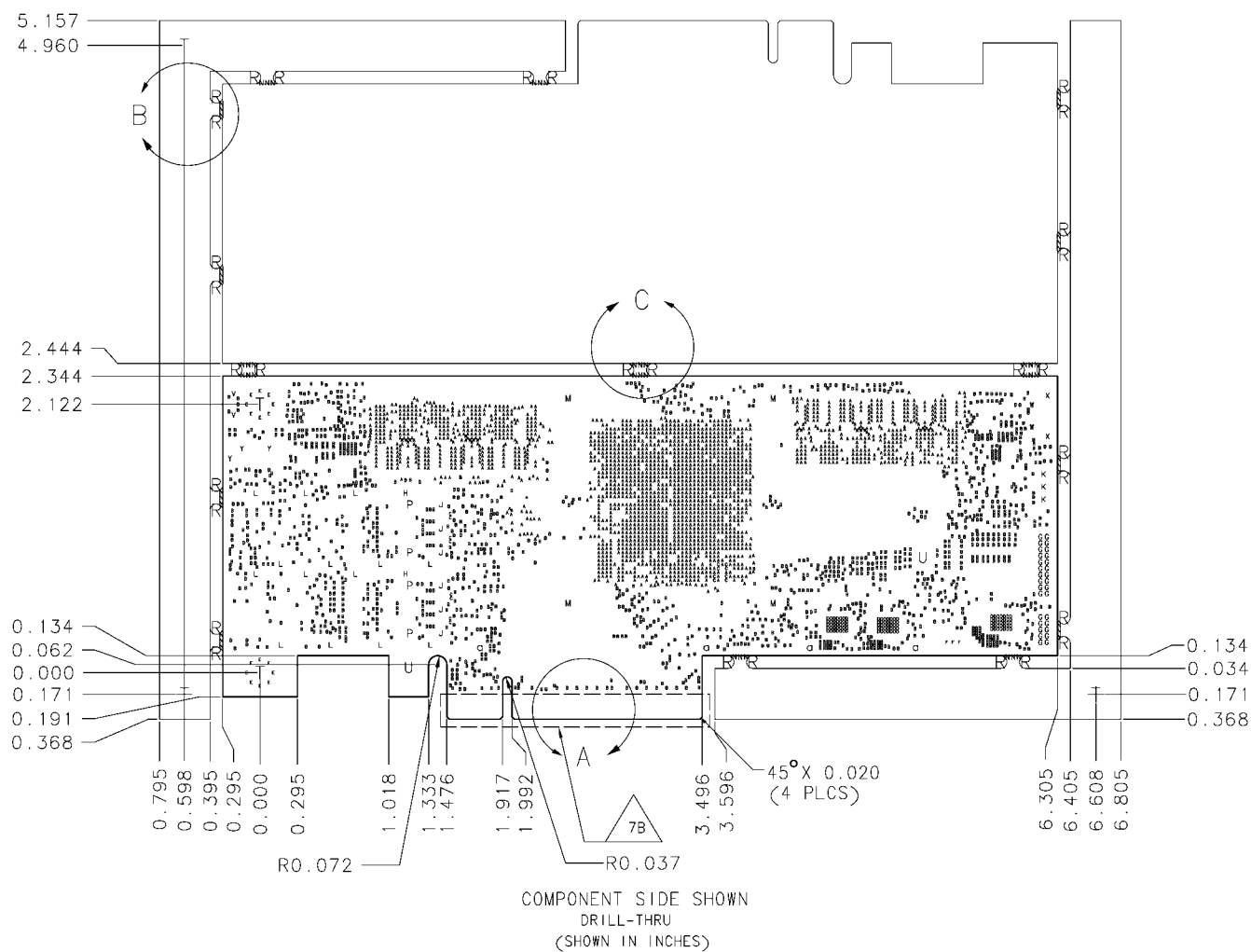


7 Physical and Electrical Specifications

7.1 Adapter Physical Dimensions

The PS225 adapters conform to the PCI Express Card Electromechanical Specification as low-profile half-length form factor. The board dimensions are shown in [Figure 3](#).

Figure 3: Board Dimensions



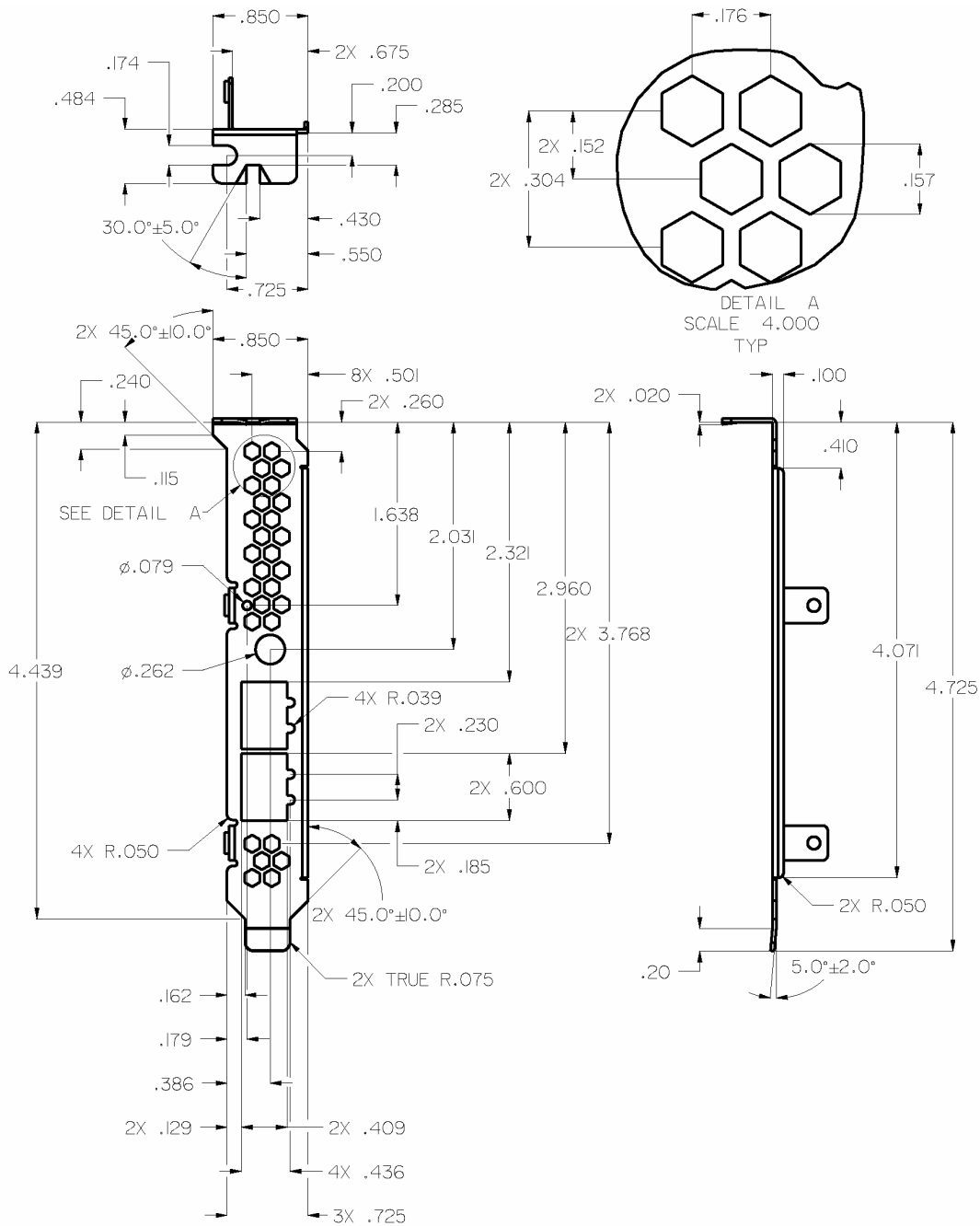
7.2 I/O Bracket Dimensions

The PS225 adapters support both standard-profile and low-profile I/O brackets.

7.2.1 Standard-Profile Bracket Dimensions

A standard-profile bracket is shown in [Figure 4](#).

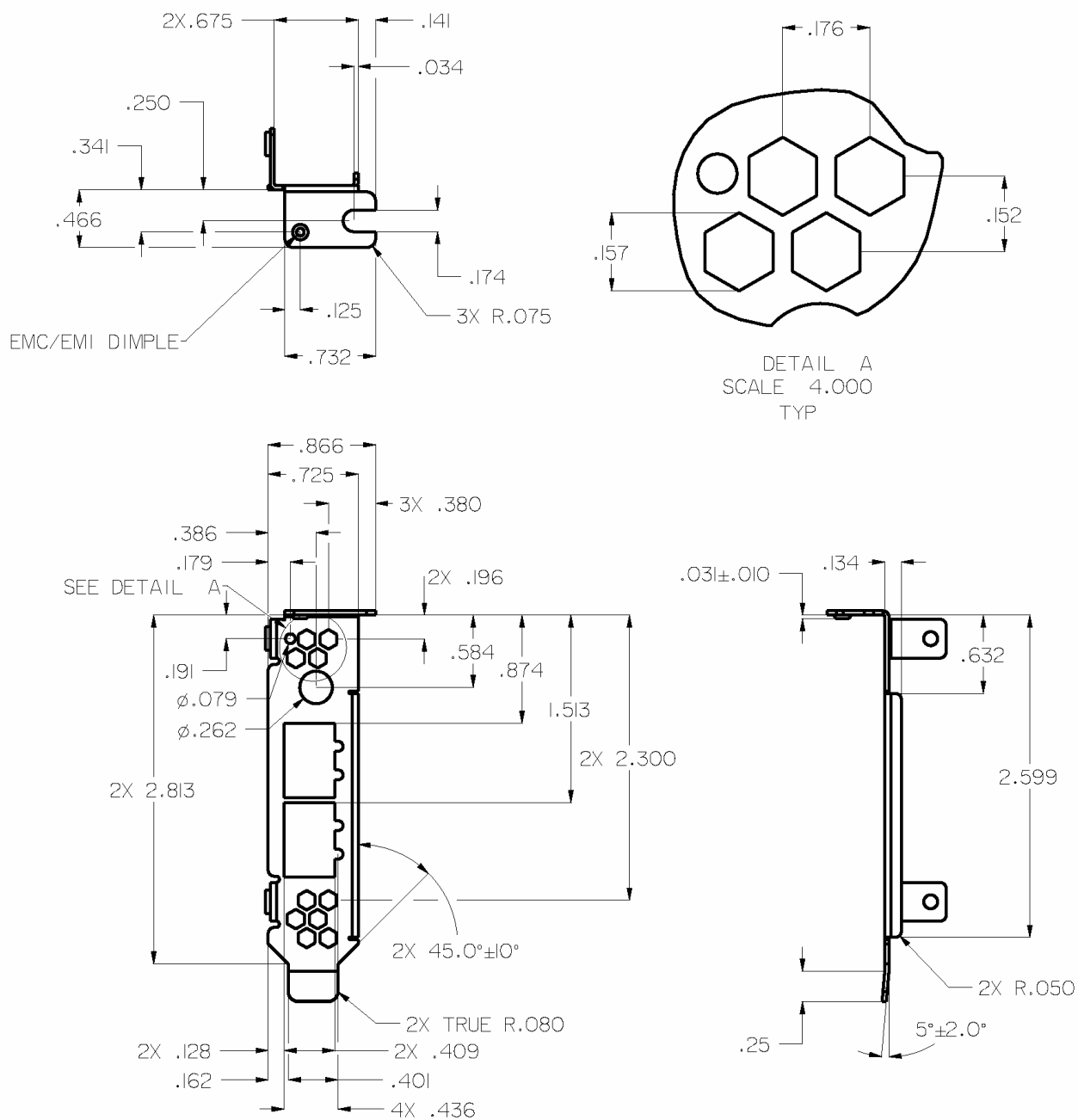
Figure 4: Standard-Profile Bracket Dimensions



7.2.2 Low-Profile Bracket Dimensions

A low-profile bracket is shown in [Figure 5](#).

Figure 5: Low-Profile Bracket Dimensions



7.3 Adapter Power

The PS225 adapters have typical power consumption of 36W in full operation under nominal environmental conditions of 290 LFM (1.47 m/s) airflow at 50°C. The maximum power consumption is 40W under worst-case environmental conditions of 100 LFM (0.51 m/s) airflow at 50°C.

NOTE: Adapter power estimates are based on passive direct-attach-copper (DAC) cables. Use of active modules will increase total board power by the amount consumed by the modules.

7.4 Adapter Thermal Design

7.4.1 Operating Environment

The recommended system environment for the PS225 adapters is 290 LFM (1.47 m/s) airflow at 50°C.

7.4.2 Heatsink

The PS225 adapters use an active heatsink (fan-sink) to ensure reliable operation in the recommended operating environment.

7.4.3 Thermal Reporting

The BCM58802H device has on-die thermal diodes that are used by software to monitor the temperature of the device.

7.5 Environment Specifications

Table 4: Environment Specifications

Parameter	Condition
Storage Temperature	–40°C to +70°C
Storage Humidity	5% to 95% non-condensing
Vibration and Shock	IEC78-2-(*) and IEC721-3-(*)

8 Regulatory and Safety

8.1 Regulatory

Table 5: Regulatory Approvals

Item	Applicable Standard	Approval (A)/Certificate (C)
CE/European Union	EN 62368-1:2014	CB report and certificate
UL/USA	IEC 62368-1 (ed. 2)	CB report and certificate

8.2 Safety

The PS225 adapters will undergo the following testing for product safety.

Table 6: Safety Approvals

Country	Certification Type/Standard	Compliance
International	CB Scheme ICES 003 – Digital Device UL 1977 (connector safety) UL 796 (PCB wiring safety) UL 94 (flammability of parts)	Yes

8.3 Electromagnetic Compatibility (EMC)

The PS225 adapters will undergo the following testing for Electromagnetic Compatibility (EMC).

Table 7: Electromagnetic Compatibility

Standard/Country	Certification Type	Compliance
CE/EU	EN 55032:2012/AC:2013 Class B EN 55024:2010 EN 61000-3-2:2014 EN 61000-3-3:2013	CE report and CE DoC
FCC/USA	CFR47 Part 15 Subpart B Class B	FCC/IC DoC and EMC report referencing FCC and IC standards
IC/Canada	ICES-003 Class B	FCC/IC DoC and report referencing FCC and IC standards
ACA/Australia, New Zealand	AS/NZS CISPR 22:2009 +A1 :2010	ACA certificate
BSM/Taiwan	CNS 13438 (2006) Class B	BSMI certificate
BSMI/Taiwan	CNS 15663	BSMI certificate/RoHS table
MSIP/S. Korea	KN32 Class B KN35	Korea certificate MSIP mark
VCCI/Japan	VCCI V-3 (2015-04)	Copy of VCCI online certificate

8.4 Electrostatic Discharge (ESD) Compliance

The PS225 adapters will undergo the following testing for Electrostatic Discharge (ESD) Compliance.

Table 8: ESD Compliance Summary

Standard	Certification Type	Compliance
EN 55024:2010 (EN 61000-4-2)	Air/Direct discharge	Yes

9 Ordering Information

Table 9: Ordering Information

Version	Description	Ordering Part Number
PS225-H04	Dual-Port 25 Gb/s Ethernet PCIe SmartNIC with 4 GB memory	BCM958802A8044C
PS225-H08	Dual-Port 25 Gb/s Ethernet PCIe SmartNIC with 8 GB memory	BCM958802A8048C
PS225-H16	Dual-Port 25 Gb/s Ethernet PCIe SmartNIC with 16 GB memory	BCM958802A8021C

Revision History

PS225-HXX-DS100; June 27, 2017

Updated:

- General Updates

PS225-HXX-DS100; December 26, 2017

Initial release.

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